

GSC Timesheet Submission Module

Orbit HRMS · BLR World · Technical Implementation Plan & Engineering Quotation

CLIENT ORGANIZATION	BLR World		
PROJECT SCOPE	GSC Weekly Timesheet Submission, Master Extension & Finance Output Engine		
TARGET DELIVERY	STANDARD RATE	TOTAL ENGINEERING SCOPE	ENGINEERING LEADS
5-7 Business Days	\$25.00 / Hour (Fixed)	24.0 Hours \$600.00	Nomeshwer Sharma & Hardik Sharma

1. EXECUTIVE SUMMARY & SCOPE ACCEPTANCE

BLR World is commissioning a streamlined, automated **Weekly Timesheet Submission Module** for its Global Service Company (GSC) personnel, natively embedded within **Orbit HRMS (AppSheet)**. The objective is to provide a frictionless time-tracking experience for staff recording actual days worked against pre-approved internal activities and external projects, while establishing an audit-proof financial data pipeline that snapshots point-in-time day rates, enforces rigorous validation rules, and produces clean Google Sheets output for direct client billing and intercompany charging.

OmmNoMi Automation LLP formally accepts the requirements defined in `BLR GSC Timesheet Submission.docx` and `Timesheet Tables.xlsx`. Applying our *Reuse-First Engineering Architecture*, we design an additive, zero-disruption system that delivers the full production-ready capability in exactly **24.0 engineering hours (\$600.00)**.

24.0 hrs	\$600.00	4 Tables	100% Frozen
TOTAL QUOTED EFFORT	TOTAL COST (@ \$25/HR)	ADDITIVE ARCHITECTURE	IMMUTABLE DAY RATES

2. OPERATIONAL OBJECTIVES & ARCHITECTURAL DESIGN PRINCIPLES

The timesheet module adheres strictly to OmmNoMi enterprise engineering standards, optimizing user experience while guaranteeing downstream accounting integrity:

ENGINEERING PRINCIPLE	SYSTEM DESIGN & ARCHITECTURAL STANDARD	BUSINESS BENEFIT FOR BLR WORLD
Frictionless Submission	Auto-snapping to the current working Friday, pre-filtered activities, dynamic 5-day inline totals, and mobile-first PWA responsiveness.	Minimizes logging friction and eliminates user confusion around billing weeks.
Immutable Rate Snapshotting	Physical snapshotting of employee day rate cards and cost codes at the exact moment of submission into <code>TimesheetOutput</code> .	Past historical financial records remain 100% protected against future rate changes.
Zero-Disruption Additive Build	Non-breaking extension of the existing <code>Employee</code> master; timesheet tables exist as independent modular schemas.	Zero operational downtime or regression risk for active Orbit HRMS users.
Automated HR Governance	Live Process Owner compliance dashboard paired with automated Friday submission reminders and Monday escalation alerts.	Reduces manual HR follow-up time by over 90% each payroll cycle.

3. DEVELOPER RESPONSES TO CLIENT BRIEF (SECTION 16 CONFIRMATION)

OmmNoMi provides definitive technical specifications for the 5 key architectural decisions requested in Section 16 of the client brief:

TOPIC	OMMNOMI PROPOSED TECHNICAL ARCHITECTURE	OPERATIONAL IMPACT & BENEFIT
1. Table Architecture	Implement a Two-Tier Header-Detail Pattern : TimesheetSubmission (Parent: Week Ending, Employee, Total Days, Status). TimesheetEntry (Child: Activity, Actual Days, Notes). TimesheetOutput (Finance Archive: Denormalized flat snapshot rows). TimesheetActivity (Controlled Catalog Master).	Guarantees relational data integrity, enables real-time sum validations, prevents orphan records, and isolates financial archives from UI operations.
2. Employee Master	Extend Orbit's existing Employee table with 3 physical columns: IsBillable (Yes/No) DayRateAED (Price / Decimal, e.g. 2500, 3500, 4000) Team (Enum: Business Development, Recruitment, SET, Finance, Performance)	Zero disruption to live profiles. Controlled fields editable exclusively by HR/Finance Admins. Rates are auto-fetched and never manually typed.
3. Revisions & Voids	Audit-Safe Voiding Workflow : - Submissions are strictly locked upon submission. - Authorized reopen marks original TimesheetOutput rows as Invoice Status = "Superseded". - Resubmission (e.g. v2) captures corrected days and appends fresh finance rows.	Zero Duplicate Billing risk in Finance. Forensic historical audit trail remains 100% intact without silent overwrites or data loss.
4. Submission Lock	Editable_If rule: AND([Status] = "Draft", [EmployeeID].[Email] = USEREMAIL()). - Slices hide Edit/Delete actions once [Status] = "Submitted". TimesheetOutput table set to Read-Only / System-Append Only.	End-users cannot tamper with historical timesheets once submitted. Creates a legally robust, audit-proof billing pipeline.
5. Future GSC Split	The Entity field in TimesheetActivity is designed as an extensible controlled tag (UAE, Qatar, KSA, HoldCo, GSC).	Fully forward-compatible with BLR's upcoming entity reorganization without requiring structural database schema redesigns later.

4. APPSHEET CONFIGURATION PROTOCOL (RULE A2 STRICT SOP ADHERENCE)

Level 1: Google Sheets Level (Physical Columns)

Employee: Add **IsBillable**, **DayRateAED**, **Team**.

TimesheetActivity: **ID**, **Name**, **Description**, **CostCentre**, **Bil**
lableStatus, **Entity**, **ProjectOwner**, **Status**, System Audit.

TimesheetSubmission: **ID**, **EmployeeID**, **WeekEndingFriday**, **Su**
bmissionDate, **Status**, **TotalDaysWorked**, **Notes**, System
Audit.

TimesheetEntry: **ID**, **SubmissionID**, **ActivityID**, **ActualDaysW**
orked, **LineNotes**, System Audit.

TimesheetOutput: Exact 14 columns matching client Excel template.

Level 2: AppSheet Table Level (Structure & Modes)

Schema Regeneration: Regenerate existing **Employee** table.

New AppSheet Tables: Add **TimesheetActivity**, **TimesheetSubmiss**
ion, **TimesheetEntry**, **TimesheetOutput**.

Update Permissions:

- TimesheetSubmission**: Adds, Updates.
- TimesheetEntry**: Adds, Updates, Deletes (Parent **IsPartOf = TRU**
E).
- TimesheetOutput**: Read_Only (System action append).
- TimesheetActivity**: Updates_Only (Admin full CRUD).

Level 3: Column Type Level (Types, Refs & Rules)

Enum Refs: **TimesheetSubmission.EmployeeID** (Enum Ref → Employee), **TimesheetEntry.ActivityID** (Enum Ref → TimesheetActivity).

Friday Validation: **WEEKDAY([WeekEndingFriday]) = 6**.

Numeric Constraints: **AND([ActualDaysWorked] > 0, [ActualDaysW**
orked] <= 7).

Initial Value Formulas:

- WeekEndingFriday**: **TODAY() + (6 - WEEKDAY(TODAY()))**
- EmployeeID**: **ANY(SELECT(EmployeeID], [Email] = USEREMA**
IL()))
- InvoiceStatus**: **"Not Invoiced"**

Level 4: Virtual Column Level (Strict In-Memory)

TimesheetSubmission.CalculatedWeeklyTotal:
SUM([Related TimesheetEntrys][ActualDaysWorked])

TimesheetSubmission.DuplicateCheck:
COUNT([Related TimesheetEntrys][ID]) <> COUNT(UNIQUE([Re
lated TimesheetEntrys][ActivityID]))

TimesheetEntrv.ActivityDescription:
[ActivityID].[Description] (Hover / detail query)

Employee.IsTimesheetSubmittedThisWeek:
Dynamically computes current weekly compliance state.

5. USER INTERFACE ARCHITECTURE & COMPLIANCE COCKPIT

VIEW COMPONENT	TARGET ROLE	UX DESIGN & FUNCTIONAL BEHAVIOR
Weekly Timesheet Form	GSC Employees <code>U_Employee</code>	Dynamic entry form. Auto-snaps to current Friday. Child line table allows logging multiple activities. Real-time indicator tallies weekly days (e.g. <code>5.0 / 5.0 Days</code>). Warns if total deviates from standard week. Duplicate check blocks double-logging.
My Submissions View	GSC Employees <code>U_Employee</code>	Interactive history card list. Displays recent weeks with visual status chips: <code>Submitted</code> or <code>Missing</code> . Provides read-only breakdown of logged days and activities.
Process Owner Cockpit	Line Managers / HR <code>U_System_Admin</code>	Management compliance dashboard displaying real-time KPI summary (Expected, Submitted, Outstanding, % Compliance), interactive team filters, and one-click nudge reminder notifications.
Finance Output Grid	Finance Admins <code>U_Finance_Admin</code>	Denormalized tabular view of <code>TimesheetOutput</code> matching client finance sheet format. Displays frozen day rates, billable tags, calculated totals, and manual <code>Invoice Status</code> tracking dropdown.

6. PHASED DELIVERY PLAN & ITEMIZED HOURLY QUOTATION (\$25.00/HR)

In accordance with OmmNoMi Commercial SOP, engineering tasks are itemized by deliverable. The complete module is quoted at our standard **\$25.00 / hour** rate, fixed to exactly **24.0 Hours** total effort.

MODULE & SCOPE	DETAILED ENGINEERING TASKS & SOP DELIVERABLES	HOURS	COST (\$)
Mod 1: Database Setup	Extend Employee table; seed 25 Activity catalog records; build Submission & Entry tables with UUID keys; configure table regeneration, Enum Refs, and cascading deletes.	4.0 hrs	\$100.00
Mod 2: Entry UX & Rules	Friday ISO auto-snapping logic & rejection validation; multi-row entry form UX; daily limits (<code>> 0</code> and <code><= 7</code>); duplicate activity blocker; real-time weekly total calculator.	5.5 hrs	\$137.50
Mod 3: Finance Archive	Create 14-column <code>TimesheetOutput</code> sheet; build atomic action stack (<code>Submit</code> → <code>ArchiveLines</code> → <code>FreezeRates</code>); billing month derivation; immutability verification.	5.0 hrs	\$125.00
Mod 4: Revisions & Voids	Row-level lock expressions (<code>Editable_If</code>) sealing submitted records; authorized reopen workflow; automatic superseded voiding; versioned resubmission mechanism (v2, v3).	3.5 hrs	\$87.50
Mod 5: Compliance Cockpit	Process Owner compliance summary dashboard (Expected vs Submitted vs Missing); submission matrix; Friday 2:00 PM GST reminder bot; Monday 10:00 AM GST escalation bot.	3.5 hrs	\$87.50
Mod 6: QA, Guide & UAT	Pre-seed test scenarios across personas; author in-app documentation guide in <code>AppResources</code> ; facilitate client UAT review window and deliver implementation walkthrough.	2.5 hrs	\$62.50
TOTAL ENGINEERING SCOPE & QUOTATION		24.0 hrs	\$600.00

Fixed-Price Commercial Guarantee: The agreed engineering scope is capped at exactly **24.0 billable hours** (\$600.00 USD). Any internal calibration, regression testing, or minor formula tweaks required during the 14-day warranty period are delivered at zero additional billing to BLR World.

7. QUALITY ASSURANCE & ACCEPTANCE TEST MATRIX

Each engineering deliverable undergoes rigorous verification against live edge cases and permission boundaries prior to UAT sign-off:

SCENARIO	VERIFICATION TEST PROCEDURE	EXPECTED ACCEPTANCE RESULT
1. Friday Snapping	Select date in form; attempt saving non-Friday dates.	Non-Fridays rejected; auto-snaps to current Friday.
2. Input Validation	Enter negative/zero days or duplicate activity rows.	Form blocks saving with explicit inline error.
3. Rate Freeze Test	Submit timesheet, then change employee day rate in master.	Historical archive record remains 100% frozen.
4. Lock Enforcement	Attempt editing a submitted timesheet as employee.	Edit and delete buttons hidden; row is read-only.
5. Revision & Void	Trigger authorized reopen; submit corrected timesheet.	Old row marked "Superseded"; new row saved.

8. APPSHEET AUTOMATION BOTS & ATOMIC ACTION PIPELINES

AUTOMATION COMPONENT	TRIGGER MECHANISM	TECHNICAL IMPLEMENTATION & EXECUTION STACK
Atomic Archive Stack	Submission Save [Status] = "Submitted"	Executes atomic grouped action: (1) Set Status to Submitted → (2) Iterate child TimesheetEntry rows → (3) Append denormalized records into TimesheetOutput with snapshot DayRateAED and BillingMonth → (4) Apply permanent row lock.
Friday Reminder Bot	Weekly Scheduled Friday 2:00 PM GST	Evaluates all active billable employees where IsTimesheetSubmittedThisWeek = FALSE. Dispatches targeted in-app push notification and email reminder to submit weekly hours before cutoff.
Monday Escalation Bot	Weekly Scheduled Monday 10:00 AM GST	Evaluates unresolved missing timesheets for prior week. Dispatches consolidated summary report to Line Managers and HR Admins with one-click escalation nudges.

9. IMPLEMENTATION ROADMAP & MILESTONES (5-7 BUSINESS DAYS)

MILESTONE PHASE	ENGINEERING DELIVERABLES & TECHNICAL FOCUS	TARGET TIMELINE
Phase 1: Database Setup	Employee master extension, TimesheetActivity catalog seeding, Google Sheets schema build.	Days 1 – 2
Phase 2: UX & Rules	Dynamic submission form, auto-Friday snapping, daily limits, duplicate activity validation.	Days 3 – 4
Phase 3: Finance Engine	Append-only archive action stack, point-in-time rate freeze, revisions & voiding workflow.	Day 5
Phase 4: Cockpit & Bots	Process Owner compliance grid, Friday reminder bot, Monday escalation notification.	Day 6
Phase 5: QA & Handover	Comprehensive scenario validation, in-app documentation guide, client UAT walkthrough.	Day 7 (UAT Sign-off)

10. COMMERCIAL ENGAGEMENT TERMS & SCOPE BOUNDARIES

✓ STANDARD COMMERCIAL RATE

All engineering effort is billed at our standard rate of **\$25.00 per hour**, totaling **\$600.00** for the agreed 24.0-hour scope.

✓ MILESTONE PAYMENT STRUCTURE

50% mobilization advance (**\$300.00**) required to initiate build; balance 50% (**\$300.00**) payable upon final UAT sign-off.

✓ DELIVERY SCHEDULE & HYPERCARE

Deployment completed within **5–7 business days**. Includes **14 days of post-go-live warranty** covering calibration and minor tweaks.

✗ CONFIRMED OUT-OF-SCOPE

PDF invoice generation, accounting ERP API sync (Xero/QuickBooks), payroll calculations, and multi-tier manager routing are excluded.

11. SCOPE GOVERNANCE, SLA COMMITMENTS & COMMUNICATION

Operational SLA & Support Channels: During active development and the 14-day warranty period, OmmNoMi commits to a **< 4-hour initial response SLA** for critical blockers during standard business hours (09:00–18:00 GST). Formal changes to scope are handled via documented change requests evaluated at our fixed \$25.00/hr rate.

12. PROFESSIONAL ENGINEERING SIGNATURES & AUTHORIZATION

Nomeshwer Sharma

Business Process Developer & Implementor

OmmNoMi Automation LLP · To engineer self-sustaining, intelligent operational ecosystems that empower human creativity and eliminate friction.

Hardik Sharma

Assistant Developer · Product Research & Development

OmmNoMi Automation LLP · To build technical solutions that keep the product at the cutting edge of innovation.